MSG No. 138.529 P_C4_2/ncm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `m'.mm`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[1,6,15,16,23,24,25,30]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[2,3,12,13,20,21,26,27]`
3	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[4,5,10,11,18,19,28,29]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[7,8,9,14,17,22,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `m'.m'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,6,7,8,25,30,31,32]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,3,4,5,26,27,28,29]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,14,15,16,17,22,23,24]`
4	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[10,11,12,13,18,19,20,21]`

Table 3: Wyckoff site: `4c`, site symmetry: `m'.mm'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,7,9,15,22,24,30,32]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,5,10,13,19,20,27,28]`
3	$[0, 0, 0]$	`[3,4,11,12,18,21,26,29]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,8,14,16,17,23,25,31]`

Table 4: Wyckoff site: `4d`, site symmetry: `m'.mm'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,7,9,15,22,24,30,32]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,5,10,13,19,20,27,28]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[3,4,11,12,18,21,26,29]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[6,8,14,16,17,23,25,31]`

Table 5: Wyckoff site: `4e`, site symmetry: `-4'2'm`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,6,15,16,20,21,26,27]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,12,13,23,24,25,30]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[4,5,10,11,17,22,31,32]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,8,9,14,18,19,28,29]`

Table 6: Wyckoff site: `4f`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,6,10,11,20,21,31,32]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,9,14,23,24,28,29]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[4,5,15,16,17,22,26,27]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,8,12,13,18,19,25,30]`

Table 7: Wyckoff site: `8g`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[1,6,15,16]`
2	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[2,3,12,13]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[4,5,10,11]`
4	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[7,8,9,14]`
5	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[17,22,31,32]`
6	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[18,19,28,29]`
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[20,21,26,27]`
8	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[23,24,25,30]`

Table 8: Wyckoff site: `8h`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,6,31,32]`
2	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[2,3,28,29]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[4,5,26,27]`
4	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[7,8,25,30]`
5	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[9,14,23,24]`
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[10,11,20,21]`
7	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[12,13,18,19]`
8	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[15,16,17,22]`

Table 9: Wyckoff site: `8i`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,15,22,32]`
2	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[2,13,19,28]`
3	$\left[0, 0, z+\frac{1}{2}\right]$	`[3,12,18,29]`
4	$\left[0, 0, \frac{1}{2}-z\right]$	`[4,11,21,26]`
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	`[5,10,20,27]`
6	$\left[0, \frac{1}{2}, z\right]$	`[6,16,17,31]`
7	$\left[\frac{1}{2}, 0, -z\right]$	`[7,9,24,30]`
8	$\left[0, \frac{1}{2}, -z\right]$	`[8,14,23,25]`

Table 10: Wyckoff site: `8j`, site symmetry: `m'.2'm`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	`[1,16,23,30]`
2	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[2,12,21,27]`
3	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	`[3,13,20,26]`
4	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[4,10,19,29]`
5	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	`[5,11,18,28]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[6,15,24,25]`
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,14,17,32]`
8	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[8,9,22,31]`

Table 11: Wyckoff site: `8k`, site symmetry: `m'.2m'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,7,30,32]`
2	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[2,5,27,28]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,4,26,29]`
4	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	`[6,8,25,31]`
5	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[9,15,22,24]`
6	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[10,13,19,20]`
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[11,12,18,21]`
8	$\left[x+\frac{1}{2},\ x,\ 0\right]$	`[14,16,17,23]`

Table 12: Wyckoff site: `8l`, site symmetry: `m'.2'm`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,16,23,30]`
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}\right]$	`[2,12,21,27]`
3	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,13,20,26]`
4	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[4,10,19,29]`
5	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,11,18,28]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0\right]$	`[6,15,24,25]`
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[7,14,17,32]`
8	$[-x,\ -x,\ 0]$	`[8,9,22,31]`

Table 13: Wyckoff site: `8m`, site symmetry: `m'.2m'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[1,7,30,32]`
2	$[-x,\ x,\ 0]$	`[2,5,27,28]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[3,4,26,29]`
4	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	`[6,8,25,31]`
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[9,15,22,24]`
6	$[x,\ -x,\ 0]$	`[10,13,19,20]`
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[11,12,18,21]`

continued ...

Table 13

No.	position	mapping
8	$\left[x+\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[14,16,17,23]

Table 14: Wyckoff site: 16n, site symmetry: .2'.

No.	position	mapping
1	$\left[\frac{3}{4},\, y,\, \frac{3}{4}\right]$	[1,21]
2	$\left[\frac{1}{2}-y,\, \frac{3}{4},\, \frac{1}{4}\right]$	[2,24]
3	$\left[y,\, \frac{3}{4},\, \frac{1}{4}\right]$	[3,23]
4	$\left[\frac{1}{4},\, -y,\, \frac{3}{4}\right]$	[4,22]
5	$\left[\frac{1}{4},\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[5,17]
6	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	[6,20]
7	$\left[y+\frac{1}{2},\, \frac{1}{4},\, \frac{1}{4}\right]$	[7,19]
8	$\left[-y,\, \frac{1}{4},\, \frac{1}{4}\right]$	[8,18]
9	$\left[\frac{1}{4},\, -y,\, \frac{1}{4}\right]$	[9,29]
10	$\left[y+\frac{1}{2},\, \frac{1}{4},\, \frac{3}{4}\right]$	[10,32]
11	$\left[-y,\, \frac{1}{4},\, \frac{3}{4}\right]$	[11,31]
12	$\left[\frac{3}{4},\, y,\, \frac{1}{4}\right]$	[12,30]
13	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[13,25]
14	$\left[\frac{1}{4},\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[14,28]
15	$\left[\frac{1}{2}-y,\, \frac{3}{4},\, \frac{3}{4}\right]$	[15,27]
16	$\left[y,\, \frac{3}{4},\, \frac{3}{4}\right]$	[16,26]

Table 15: Wyckoff site: 16o, site symmetry: ..m

No.	position	mapping
1	$\left[x,\, \frac{1}{2}-x,\, z\right]$	[1,15]
2	$\left[x,\, x,\, z+\frac{1}{2}\right]$	[2,13]
3	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[3,12]
4	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[4,11]
5	$\left[-x,\, -x,\, \frac{1}{2}-z\right]$	[5,10]
6	$\left[\frac{1}{2}-x,\, x,\, z\right]$	[6,16]
7	$\left[-x,\, x+\frac{1}{2},\, -z\right]$	[7,9]
8	$\left[x+\frac{1}{2},\, -x,\, -z\right]$	[8,14]
9	$\left[x+\frac{1}{2},\, -x,\, z\right]$	[17,31]
10	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[18,29]
11	$\left[-x,\, -x,\, z+\frac{1}{2}\right]$	[19,28]
12	$\left[x,\, x,\, \frac{1}{2}-z\right]$	[20,27]
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[21,26]
14	$\left[-x,\, x+\frac{1}{2},\, z\right]$	[22,32]
15	$\left[\frac{1}{2}-x,\, x,\, -z\right]$	[23,25]
16	$\left[x,\, \frac{1}{2}-x,\, -z\right]$	[24,30]

Table 16: Wyckoff site: `16p`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ -x,\ z]$	`[1,31]`
2	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	`[2,29]`
3	$\left[-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[3,28]`
4	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	`[4,27]`
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[5,26]`
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	`[6,32]`
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -z\right]$	`[7,25]`
8	$[x,\ -x,\ -z]$	`[8,30]`
9	$[-x,\ x,\ -z]$	`[9,23]`
10	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	`[10,21]`
11	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,20]`
12	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	`[12,19]`
13	$\left[x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[13,18]`
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	`[14,24]`
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	`[15,17]`
16	$[-x,\ x,\ z]$	`[16,22]`

Table 17: Wyckoff site: `16q`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	`[1,30]`
2	$\left[\frac{1}{2}-y,\ x,\ 0\right]$	`[2,27]`
3	$\left[y,\ \frac{1}{2}-x,\ 0\right]$	`[3,26]`
4	$\left[x+\frac{1}{2},\ -y,\ 0\right]$	`[4,29]`
5	$\left[-x,\ y+\frac{1}{2},\ 0\right]$	`[5,28]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[6,25]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,32]`
8	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[8,31]`
9	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[9,22]`
10	$\left[y+\frac{1}{2},\ -x,\ 0\right]$	`[10,19]`
11	$\left[-y,\ x+\frac{1}{2},\ 0\right]$	`[11,18]`
12	$\left[\frac{1}{2}-x,\ y,\ 0\right]$	`[12,21]`
13	$\left[x,\ \frac{1}{2}-y,\ 0\right]$	`[13,20]`
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[14,17]`
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[15,24]`
16	$\left[y,\ x,\ \frac{1}{2}\right]$	`[16,23]`

Table 18: Wyckoff site: `32r`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	`[4]`

continued ...

Table 18

No.	position	mapping
5	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[5]
6	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[6]
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[7]
8	$[-y,\, -x,\, -z]$	[8]
9	$[-x,\, -y,\, -z]$	[9]
10	$\left[y+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$	[10]
11	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[13]
14	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[14]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[15]
16	$[y,\, x,\, z]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[17]
18	$\left[-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\, -x,\, z+\frac{1}{2}\right]$	[19]
20	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	[21]
22	$[-x,\, -y,\, z]$	[22]
23	$[y,\, x,\, -z]$	[23]
24	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[25]
26	$\left[y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x,\, \frac{1}{2}-z\right]$	[27]
28	$\left[-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[28]
29	$\left[x+\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	[29]
30	$[x,\, y,\, -z]$	[30]
31	$[-y,\, -x,\, z]$	[31]
32	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[32]